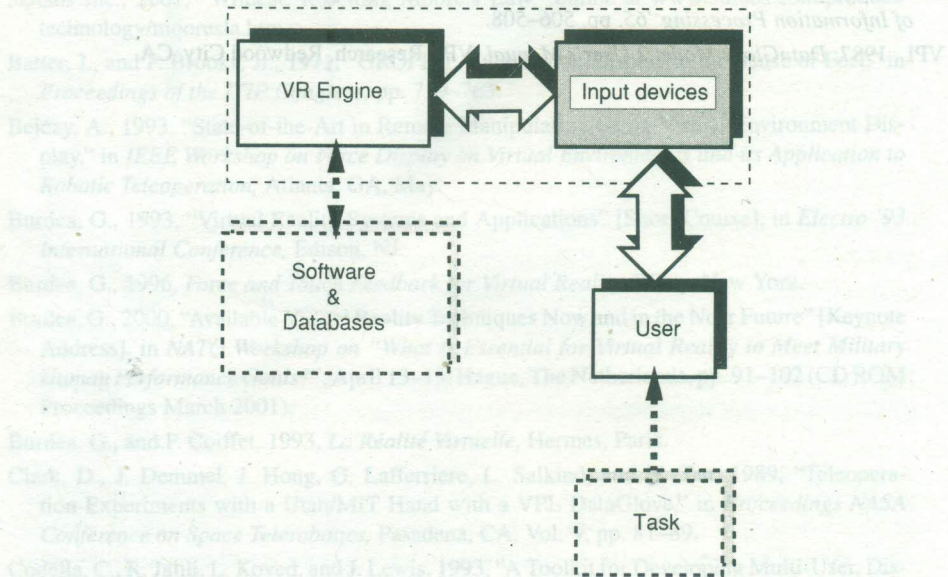


CHAPTER 2

INPUT DEVICES: TRACKERS, NAVIGATION, AND GESTURE INTERFACES

VR System Architecture



One of the three I's defining virtual reality stands for *interactivity*. In order to allow human-computer interaction it is necessary to use special interfaces designed to input a user's commands into the computer and to provide feedback from the simulation to the user. Today's VR interfaces are varied in functionality and purpose, as they address several human sensorial channels. For example, body motion is measured with 3D position trackers or using sensing suits, hand gestures are digitized by sensing gloves, visual feedback is sent to stereo HMDs and large-volume displays, virtual sound is computed by 3D sound generators, etc. Some of these input/output (I/O) devices are commercially available, some are still prototypes in a field which has become a very active research area. The aim of researchers is to allow faster and more natural ways of interaction with the computer and thus overcome the communication bottleneck presented by the keyboard and the computer mouse.

This chapter describes the VR interfaces used in tracking, VR navigation, and gesture input. Output devices for visual, auditory, and haptic feedback to the user are

the focus of the next chapter. Our discussion of VR interface hardware includes functionality, design, and limitations, comparing, whenever possible, alternative available technologies.

2.1 THREE-DIMENSIONAL POSITION TRACKERS — ①

Many computer application domains, such as navigation, ballistic missile tracking, ubiquitous computing, robotics, biomechanics, architecture, computer-aided design (CAD), education, and VR, require knowledge of the real-time position and orientation of moving objects within some frame of reference [Hightower and Borriello, 2001]. These applications have varying requirements in terms of such parameters as measurement range, precision, and temporal update rates. Global positioning system (GPS) transducers, for example, report position with an accuracy of 1–5 m (95% of the time) on a worldwide scale. This is great for car navigation, but VR has much more stringent accuracy requirements. On the other hand, the scale of measured distances for VR interactions is usually not larger than a room or a laboratory.

A moving object in 3D space has six degrees of freedom, three for translations and three for rotations. If a Cartesian coordinate system is attached to the moving object (as illustrated in Fig. 2.1), then its translations are along the X , Y , and Z axes. Object rotations about these axes are called yaw, pitch, and roll, respectively. These define a dataset of six numbers that need to be measured sufficiently rapidly, as the object may be moving at high speed.

Definition The special-purpose hardware used in VR to measure the real-time change in a 3D object position and orientation is called a tracker.

Virtual reality applications typically measure the motion of the user's head, hands, or limbs for the purpose of view control, locomotion, and object manipulation [Foxlin,

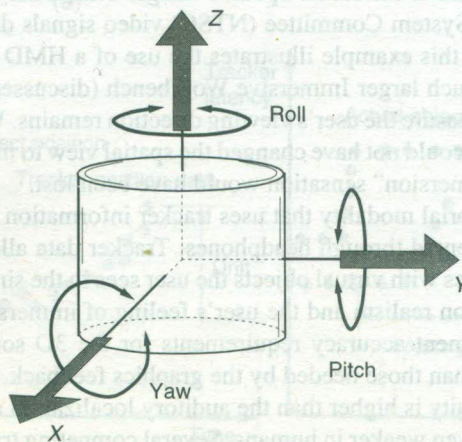


Fig. 2.1 System of coordinates of a moving 3D object. From Burdea and Coiffet [1993].
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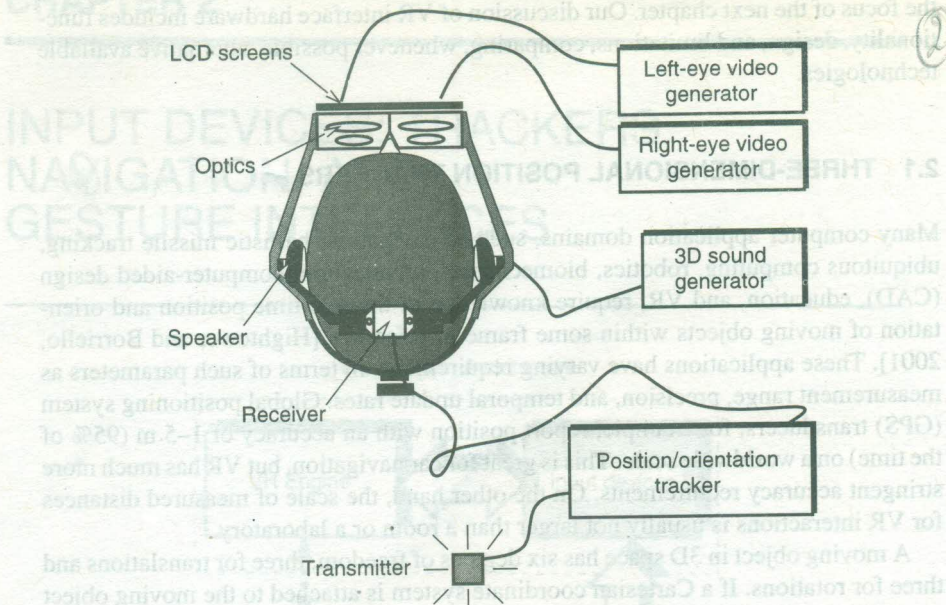


Fig. 2.2 A 3D magnetic sensor in a HMD. Adapted from Pimentel and Teixeira [1993]. Reprinted by permission.

2002]. A newer tracker application in VR is for the control of an avatar, or virtual body, mapped to the user. In the case of the head-mounted display illustrated in Figure 2.2, the tracker receiver is placed on the user's head, so that when the posture of the head changes, so does the position of the receiver. The user's head motion is sampled by an electronic unit and sent to a host computer (in this case a graphics workstation). The computer uses the tracker data to calculate a new viewing direction of the virtual scene and to render an updated image. This scene is then converted to National Television System Committee (NTSC) video signals displayed by the two LCD screens. While this example illustrates the use of a HMD tracker, the display could have been a much larger Immersive Workbench (discussed in Chapter 3), but the requirement to measure the user's viewing direction remains. Without the 3D head tracker the computer could not have changed the spatial view to match the user's head posture, and the "immersion" sensation would have been lost.

Another VR sensorial modality that uses tracker information is 3D sound, which in Figure 2.2 is presented through headphones. Tracker data allow the computer to collocate sound sources with virtual objects the user sees in the simulation. This helps increase the simulation realism and the user's feeling of immersion in the synthetic world. The measurement accuracy requirements for the 3D sound application are much less stringent than those needed by the graphics feedback. As noted by Foxlin [2002], the visual acuity is higher than the auditory localization acuity, and auditory depth perception is even weaker in humans. Several competing tracking technologies are available, such as mechanical, magnetic, optical, ultrasonic, and hybrid. It is therefore necessary to look at tracker performance parameters in order to match their

measurement capabilities to different sensorial channel requirements and available budgets.

2.1.1 Tracker Performance Parameters

All 3D trackers, regardless of the technology they use, share a number of very important performance parameters, such as accuracy, jitter, drift, and latency. These are illustrated in Figure 2.3.

Definition Tracker accuracy represents the difference between the object's actual 3D position and that reported by tracker measurements.

The more accurate a tracker, the smaller this difference is and the better the simulation follows the real user's actions. Accuracy is given separately for tracking translation (fraction of a millimeter) and rotation (fraction of a degree). Accuracy is typically not constant and is degraded with distance from the origin of the reference

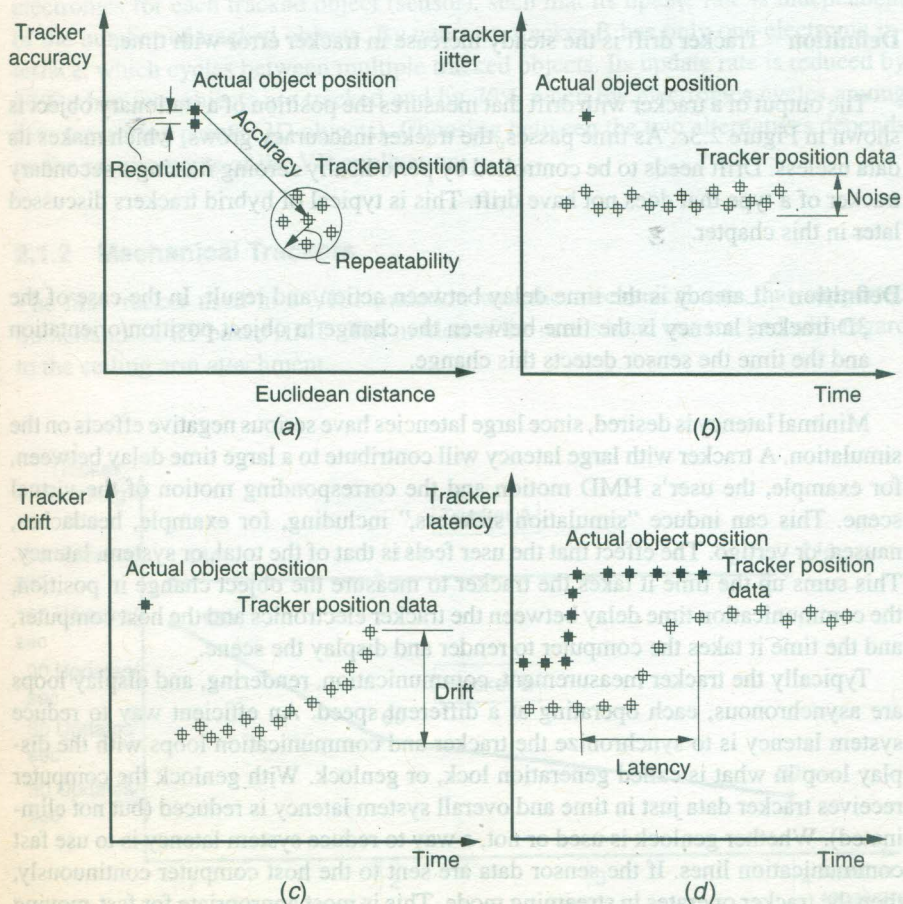


Fig. 2.3 Tracker performance parameters: (a) accuracy; (b) jitter; (c) drift; (d) latency.

2.2 NAVIGATION AND MANIPULATION INTERFACES —

③

Definition A navigation/manipulation interface is a device that allows the interactive change of the view to the virtual environment and exploration through the selection and manipulation of a virtual object of interest.

The navigation/manipulation can be done in either absolute coordinates or relative coordinates. The trackers described so far are absolute, as they return the position and orientation of a moving object with respect to a fixed system of coordinates. The position of the VR object controlled on the screen is directly mapped to the absolute position of the receiver in the world (transmitter)-fixed system of coordinates.

Another way to control a VR object's position is through relative sensors. Whereas absolute position data are never a zero set (even if the receiver is at rest), a relative position sensor will always return zeros if not acted upon. Navigation/manipulation in relative coordinates allows for incremental position control relative to the object's previous 3D position. The position of the VR object is incremented by six signed quantities at every simulation cycle. Velocity is indirectly controlled, as a larger translation/rotation increment will result in a larger velocity of the VR object in the simulation.

2.2.1 Tracker-Based Navigation/Manipulation Interfaces

Trackers offer more functionality to VR simulations than simply measuring the real-time position/orientation of the user's hand and head. Integrated within a structure that houses user-programmable pushbuttons, trackers become navigation and manipulation interfaces. Examples are the Polhemus 3Ball and the Ascension Technology 3D Mouse [Anon, 1998]. The 3Ball is a hollow billiard ball that houses a tracker receiver inside and has a pushbutton on the surface. An example of its use is to move a virtual camera that travels along the 3D vector controlled by the ball as long as the pushbutton is pressed. Another example is a wand, which projects a virtual ray along the tracker-controlled vector. Any object intersected by this ray can be selected by pushing the button on the billiard ball, and then repositioned through wrist motion. The 3D Mouse offers more functionality, as it can work both as a mouse (in 2D) and as a navigation/manipulation interface in 3D, similar to the 3Ball. Further functionality

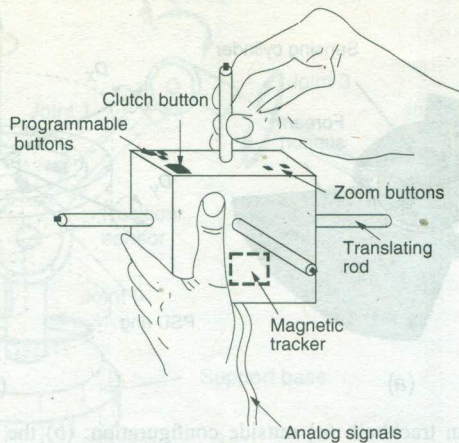


Fig. 2.21 The Cubic-Mouse. Adapted from Fröhlich et al. [2000]. © 2000 IEEE. Reprinted by permission.

is obtained through the programming of the three buttons incorporated on the 3D Mouse.

One drawback of absolute position control of virtual objects is the user's limited arm reach. The larger the dimensions of the graphics display, the more this becomes a problem. One way to solve the disparity between the user's arm reach and the (larger) display dimensions is to multiply the tracker readings with a constant (gain) larger than unity. The drawback is that tracker noise is equally amplified, and selected objects appear more jittery than they would otherwise be. An alternative is to use indexed motions, in which virtual objects are selected and deselected repeatedly. When the object of interest is deselected, it stays in place, while the user moves his or her arm back and then reselects the object and moves it using the tracker, and so on.

An example of a manipulation interface that uses this indexed motion is the Cubic-Mouse, illustrated in Figure 2.21 [Fröhlich et al., 2000]. It consists of a plastic cubic structure of dimensions $9\text{ cm} \times 9\text{ cm} \times 9\text{ cm}$ ($3\frac{5}{8}\text{ in.} \times 3\frac{5}{8}\text{ in.} \times 3\frac{5}{8}\text{ in.}$). This plastic cube houses a Polhemus Fastrack, three mutually perpendicular translating rods, and six application-programmable control buttons. The buttons are placed on the face that normally is oriented upward (the "up" face), while wires from the tracker, the sensors embedded in the translating rods, and the buttons exit through the opposite ("down") face.

As opposed to wands that are designed for flyby navigation, the Cubic-Mouse is designed primarily for the manipulation of a single, large virtual model. Navigation in this context means rotating and translating the virtual model through the tracker and dissecting/editing it using the three rods. Each of the rods controls a cutting plane, and buttons at the end of the rods allow for selecting of the orientation of the cutting planes. Further functionality is brought by the control buttons, one being dedicated to indexed motion (the clutch button) and two to zooming in-out on the virtual model. As opposed to classical computer mice, the Cubic-Mouse is designed for dual-hand operation. Users hold the mouse with their nondominant hand and push the rods and buttons with their dominant one. Proprioception (the body's own sense of position/motion) and tactile cues allow operation without looking at the rods.



VC 2.2

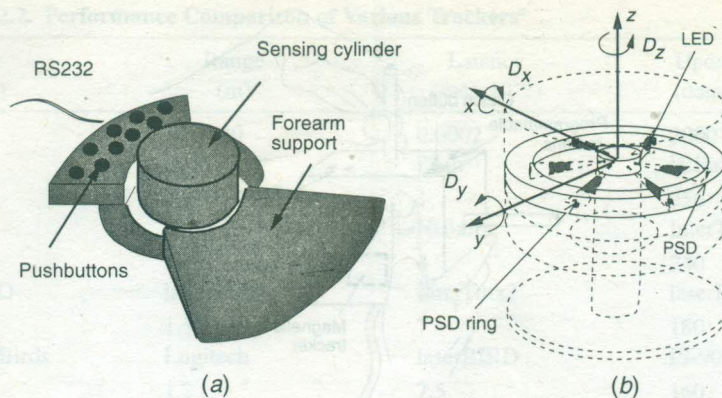


Fig. 2.22 The Magellan trackball: (a) outside configuration; (b) the sensing mechanism. Adapted from Hirzinger and Bejczy [1988]. Reprinted by permission.

2.2.2 Trackballs

A class of interfaces that allow navigation/manipulation in relative coordinates are trackballs, such as the Logitech Magellan shown in Figure 2.22a [Anon, 1998]. This is a sensorized cylinder that measures three forces and three torques applied by the user's hand on a compliant element. Forces and torques are measured indirectly based on the spring deformation law. The central part of the cylinder is fixed and has six light-emitting diodes, as shown in Figure 2.22b [Hirzinger and Bejczy, 1988]. Correspondingly, six photo sensors are placed on a moving external cylinder. When the user applies forces or torques on the moving shell, the photo sensor output is used to measure three forces and three torques. These forces and torques are then sent to a host computer over an RS232 serial line. Here they are multiplied by software gains to return a differential change in the controlled object position and orientation. Larger gains will result in larger speeds for the VR object the user controls, but its motion will not be smooth if the host cannot refresh the screen fast enough. An alternate way to control VR objects is through force control, where forces measured by the trackball are used to control forces applied by the VR object on the simulated environment. The trackball can also be used to fly by in the simulation. In that case the sensor affects the velocity and orientation of a virtual camera looking at the simulated world.

Several pushbuttons are integrated with the trackball support, within reach of the user's fingers. These buttons are binary on/off and can be programmed according to the application. For example, one button can be used to increase the simulation gains (or VR object speed), and another button to couple and decouple the trackball with the VR object it controls. Finally, some buttons may be used to start and stop the simulation, or to reset the simulation to a default start location, if the user becomes disoriented.

Trackballs suffer from sensor coupling. Although the user may wish to have the VR object translate, but not rotate, the VR object may do both translations and rotations. This is due to nonzero sensed torques when pure forces are applied by the user's fingers on the cylinder. These unwanted motions can be suppressed with software filters that

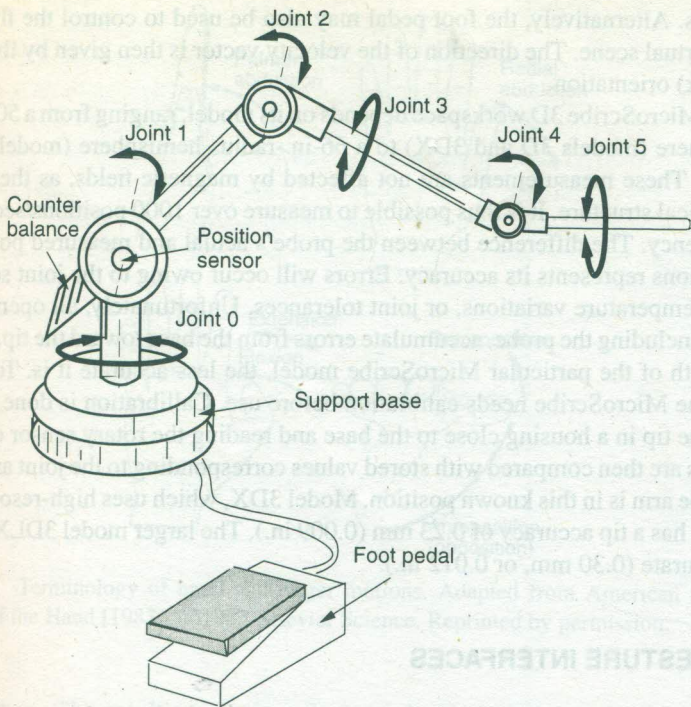


Fig. 2.23 The Immersion Co. Microscribe 3D. Adapted from Rosenberg et al. [2000]. © 2000 Immersion Co. Reprinted by permission.

only read forces and not torques, or by hardware filters. These hardware filters are pushbuttons (available on some trackballs) that allow users to select translation-only, rotation-only, or dominant-motion input.

2.2.3 Three-Dimensional Probes

Users felt a need for an I/O device that would be intuitive to use, inexpensive, and allow either absolute or relative position control of the simulation. One such device is the Immersion Probe produced by Immersion Co. in the early 1990s. This was later renamed the MicroScribe 3D and its use extended to include digitizing objects [Rosenberg et al., 2000]. It consists of a small, sensorized mechanical arm that sits on a support base, with a small 6 in. $\times$ 6 in. footprint. The probe has six joints (joints 0–5), as illustrated in Figure 2.23.

Each rotary joint represents one degree of freedom, and thus the probe has six degrees of freedom, allowing simultaneous positioning and orienting of its tip. A counterbalance is placed close to the base to minimize the user's fatigue. The tip position relative to the base is obtained through direct kinematics calculations, based on sensor values and the length of the links. Software on the host computer reads the joint sensors on an RS232 line, then uses its kinematic model to determine where the tip is. A binary switch on a foot pedal is used to select/deselect (release) virtual objects, navigate (start/stop), or mark a point on the real object surface for digitization

purposes. Alternatively, the foot pedal may also be used to control the flying speed in the virtual scene. The direction of the velocity vector is then given by the probe tip (last link) orientation.

The MicroScribe 3D workspace depends on its model, ranging from a 50-in.-radius hemisphere (models 3D and 3DX) to a 66-in.-radius hemisphere (models 3DL and 3DLX). These measurements are not affected by magnetic fields, as the probe is a mechanical structure. It is thus possible to measure over 1000 positions/sec with minimal latency. The difference between the probe's actual and measured positions and orientations represents its accuracy. Errors will occur owing to the joint sensor resolution, temperature variations, or joint tolerances. Unfortunately, all open kinematic chains, including the probe, accumulate errors from the base toward the tip. The larger the length of the particular MicroScribe model, the less accurate it is. To minimize errors the MicroScribe needs calibration before use. Calibration is done by placing the probe tip in a housing close to the base and reading the rotary sensor data. These readings are then compared with stored values corresponding to the joint angles when the probe arm is in this known position. Model 3DX, which uses high-resolution joint sensors, has a tip accuracy of 0.23 mm (0.009 in.). The larger model 3DLX is slightly less accurate (0.30 mm, or 0.012 in.).

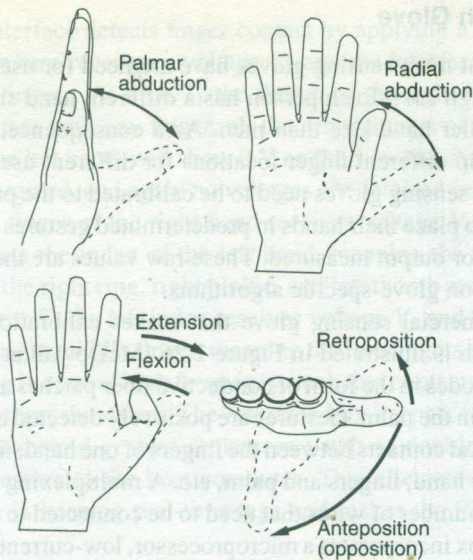
2.3 GESTURE INTERFACES

Trackballs and 3D probes have the advantage of simplicity, compactness, and quiet operation. By their nature they limit the freedom of motion of the user's hand to a small area close to the desk. Therefore the natural motion of the user's hand is sacrificed, and interaction with the virtual world is less intuitive. In order to have large gesture-based interactions with the VR simulation, it is necessary that the I/O devices maintain the hand freedom of motion within a certain volume. It is also desirable to allow additional degrees of freedom by sensing individual finger motions. The human fingers have degrees of freedom associated with flexion–extension and lateral abduction–adduction, as illustrated in Figure 2.24 [American Society for Surgery of the Hand, 1983]. Additionally, the thumb has an anteposition–retroposition motion, which brings it in opposition to the palm.

Definition Gesture interfaces are devices that measure the real-time position of the user's fingers (and sometimes wrist) in order to allow natural, gesture-recognition-based interaction with the virtual environment.

Most gesture interfaces today are sensing gloves that have embedded sensors which measure the position of each finger versus the palm. Sensing gloves differ in such factors as, for example, the type of sensors they use, the number of sensors for each finger (one or several), their sensor resolution, the glove sampling rate, and whether they are tethered or wireless.

Some of the available commercial sensing gloves are the Fakespace Pinch Glove [McDowall et al., 2000], the Fifth Dimension Technology 5DT Data Glove [Fifth Dimension, 2000], the Didjiglove [Anon, 2000], and the Immersion CyberGlove [Immersion Co., 2001]. They have sensors that measure some (or all) of the finger joint angles. Some have built-in trackers as well, in order to measure the user's



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Fig. 2.24 Terminology of hand and finger motions. Adapted from American Society for Surgery of the Hand [1983]. ©1983 Elsevier Science. Reprinted by permission.

wrist motion. The resulting sensing glove work envelope is much larger than that of trackballs or joysticks, as illustrated in Figure 2.25. As opposed to trackballs and 3D probes, which have single-point interaction with the virtual environment, sensing gloves allow dextrous, multipoint interaction at the fingertips or palm. This results in a more realistic simulation, especially for object manipulation tasks. Additionally, sensing gloves can become navigation interfaces, based on user-programmed gesture libraries [Laviola, 2000].

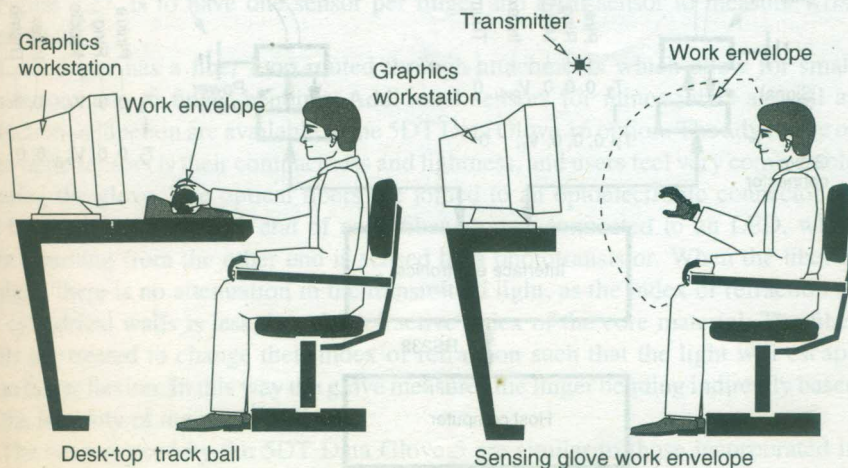


Fig. 2.25 Comparison of sensing glove work envelope and trackball work envelope. From Burdea [1993]. Reprinted by permission.